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A Three-Tier Knowledge Graph RAG for Vietnamese Legal Question Answering

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Abstract. Accurate legal question answering demands navigating hierarchical document structures, following cross-references, and multi-hop reasoning, capabilities that standard Retrieval-Augmented Generation (RAG) systems lack. Vietnamese legal text compounds this challenge through tonal ambiguity, domain-specific abbreviations, and rigid structural hierarchies. This paper presents a three-tier knowledge graph RAG architecture that integrates a domain ontology, a knowledge graph, and a document tree forest. Three retrieval tiers, LLM-guided tree traversal, dual-level semantic search with intent-aware Personalized PageRank, and Reciprocal Rank Fusion with knowledge graph expansion, operate over these representations. Evaluation on 530 expert-annotated questions across enterprise law, traffic safety law, and postgraduate education regulation shows that the proposed system retrieves at least one correct article in the top 10 results for 92.2% of queries, outperforming the strongest baseline by 7.4%. These results hold across all three domains without domain-specific tuning. To the best of the authors' knowledge, no prior work combines ontology-enhanced knowledge graphs with multi-tier retrieval for Vietnamese legal question answering.

Keywords. Knowledge Graph, Legal Question Answering, Retrieval Augmented Generation, Ontology, Vietnamese NLP, Multi-Tier Retrieval, Multi-Domain

1. Introduction

Accurate access to legal information is critical for citizens, businesses, and legal practitioners navigating regulatory compliance. Legal forums and social networks in Vietnam are flooded with questions on business formation, traffic penalties, and regulatory compliance, yet answers are often fragmented, contradictory, or cite repealed provisions.

Legal documents exhibit complex hierarchical structures: a single law organizes content into parts, chapters, sections, articles, clauses, and points, where each level carries distinct legal scope. They also contain pervasive cross-references: answering "If a member borrows money to meet charter capital and later withdraws it, what are the penalties?" requires chaining Article 47 (member obligations) to Article 16, Clause 5 (prohibited acts) across different chapters. Vietnamese compounds this challenge through tonal

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ambiguity, domain-specific abbreviations (*HDQT*, *CTCP*), and a rigid legal hierarchy (Constitution, Codes, Decrees, Circulars).

Standard RAG [1] retrieves by vector similarity [2], an approach effective in many domains [3] but inadequate for legal text: embeddings blur precise legal meanings (semantic gap), flat retrieval ignores document hierarchy (no structural awareness), and dense search cannot chain cross-references (no multi-hop reasoning). LightRAG [4] and GraphRAG [5] model entity relationships but disregard document hierarchy; RAPTOR [6] builds summary trees via recursive clustering but relies on embedding-only retrieval; PageIndex [7] exploits structure but lacks semantic reasoning; legal ontology systems [8,9] formalize domain knowledge but remain disconnected from retrieval pipelines. To address these limitations, a three-tier knowledge graph RAG is proposed that unifies structural navigation, semantic retrieval, and rank fusion: Tier 1 performs LLM-guided tree traversal, Tier 2 applies dual-level semantic retrieval with intent-aware Personalized PageRank, and Tier 3 merges outputs through Reciprocal Rank Fusion with knowledge graph expansion.

This paper makes three contributions. First, a data engineering pipeline is developed that parses Vietnamese legal documents into hierarchical trees, constructs a domain ontology, and builds a knowledge graph through semi-automatic LLM extraction with expert refinement. Second, a three-tier RAG architecture is designed that unifies structural tree traversal, semantic graph retrieval, and reciprocal rank fusion into a single pipeline. Third, a multi-intent Personalized PageRank algorithm is introduced that adapts edge weights to detected query intent, allowing the graph to emphasize different relation types for penalty, procedural, or definitional questions. Experiments on 530 questions across three domains show that the system reaches 92.2% top-10 hit rate on legal QA, surpassing PageIndex by 7.4% and RAPTOR by 51.2%, and 88.5% on education regulation, exceeding PageIndex by 13.9%.

2. Related Work

From lexical to neural retrieval. Early legal retrieval relied on lexical matching: BM25 [10] and TF-IDF [11] score documents by term overlap but fail when users paraphrase legal concepts. Dense retrieval shifted to learned representations with Sentence-BERT [2] and DPR [12]; subsequent work added self-reflection [13] and hypothetical document generation [14] (surveyed by Fan et al. [3]). Domain-adapted models such as LEGAL-BERT [15] capture legal phrasing yet still operate on flat passages. For Vietnamese, PhoBERT [16] and VnCoreNLP [17] provide foundational models, and Ngo et al. [18] combined BM25 with BERT [19] for the ALQAC 2021 competition. A shared limitation persists across these approaches: treating documents as flat collections discards the hierarchical structure critical for legal text.

Knowledge-enhanced retrieval. To recover structural and relational information, researchers turned to knowledge representations. Legal-Onto [8] combines the Rela-model [20] with conceptual graphs for Vietnamese Land Law but relies on manual rules that do not scale. GraphRAG [5] partitions entity graphs into communities with LLM summaries, and LightRAG [4] combines vector search with graph traversal. Both enable multi-hop reasoning yet treat documents as bags of entities without preserving hierarchy; in our experiments, LightRAG's entity extraction quality was low (74% of entities appeared only once). RAG [1] grounds generation in retrieved evidence but remains bottlenecked by retrieval quality. Surveys [21,22] confirm that bridging legal knowledge with scalable retrieval remains open.

Structure-aware hierarchical retrieval. RAPTOR [6] builds summary trees via recursive clustering and LLM summarization, capturing multi-granularity themes; however, it discards original document hierarchy and relies on embedding-only retrieval at query time, achieving only 41.0% top-10 hit rate on our structured legal benchmark. PageIndex [7] takes a different approach by using LLM-guided tree traversal for hierarchical document search, achieving high precision when the model selects the correct branch but missing content outside the traversal path. Crucially, neither system integrates knowledge graphs for cross-document reasoning, leaving multi-hop queries that span different laws or decrees unresolved.

Positioning. No existing system combines ontological knowledge, graph-based retrieval, and structural tree navigation for Vietnamese legal text. Neural approaches achieve broad coverage but lack structure; symbolic systems enable reasoning but lack scalability; LLM-based systems provide language understanding but hallucinate without domain grounding. Our system integrates all three capabilities through a three-tier architecture with intent-aware reasoning and rank fusion.

3. Methodology

3.1. Problem Formulation

Vietnamese legal documents are subdivided into chapters, sections, articles, clauses, and points, each level nested within the one above, forming a six-level hierarchy. Each document can be modeled as a rooted tree whose nodes correspond to these structural levels and whose edges represent parent–child containment, with articles as the retrieval target; the full corpus forms a *document forest* $\mathbf{F} = \{T_1, \ldots, T_m\}$. Every node is assigned a composite identifier encoding its position in the hierarchy (e.g., `59-2020-QH14:c2` for Chapter 2, `59-2020-QH14:d5` for Article 5), enabling unambiguous cross-tree references. Cross-reference edges link articles across trees whenever one article explicitly cites another. Given a user query q, the system must return a ranked list of the k most relevant articles from $\mathbf{F}$. Article-level granularity is essential because Vietnamese legal practice requires citing specific articles, not arbitrary text passages. Unlike standard passage retrieval over flat collections, the search space here is hierarchical, a single query may demand multi-hop reasoning across articles in different chapters or documents linked by cross-references, and Vietnamese compounds this with tonal ambiguity, domain abbreviations (*TNHH*, *CTCP*), and synonymous legal phrasing that general-purpose embeddings fail to distinguish.

3.2. System Overview

The architecture separates into an offline phase and an online phase (Figure 1). The offline phase parses legal documents into hierarchical trees, extracts entities and relations into a knowledge graph, and generates summaries for each structural level. The online phase processes each query through four stages: query analysis (Stage 0), LLM-guided tree traversal (Tier 1), dual-level semantic retrieval over the knowledge graph (Tier 2), and fusion with knowledge graph expansion (Tier 3).

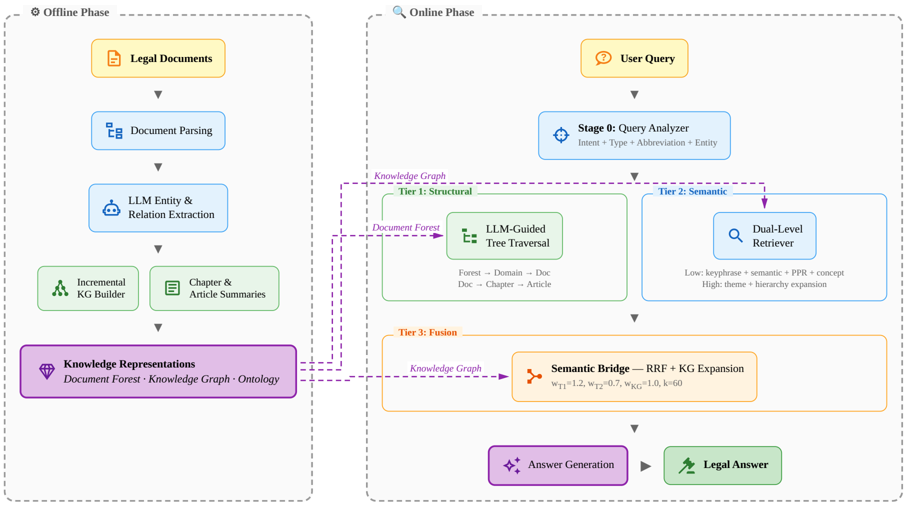

Figure 1. Architecture of the proposed three-tier retrieval system. Offline phase (left): knowledge model construction from legal documents. Online phase (right): query processing through Tier 1 (structural tree traversal), Tier 2 (dual-level semantic retrieval), and Tier 3 (RRF fusion with KG expansion).

3.3. *Data Engineering and Knowledge Construction*

The offline phase builds three knowledge representations from the raw corpus, following knowledge engineering practice [9]. First, each document is parsed into a rooted tree, forming a document forest with cross-reference edges linking articles across documents. Second, a knowledge graph is constructed via LLM extraction with expert-in-the-loop refinement. Third, chapter-level and article-level summaries are generated for LLM-guided branch selection during retrieval. Corpus statistics are reported in Table 1; implementation details in Section 4.

3.4. *Hybrid Retrieval Architecture*

3.4.1. *Stage 0: Query Analysis*

Given a raw user query q, the query analyzer first classifies its intent into one of six types (Penalty, Definition, Procedure, Requirement, Reference, General) via Vietnamese regex with LLM fallback, then expands abbreviations (e.g., *TNHH* $\rightarrow$ *trach nhiem huu han*) and extracts seed entities such as article references, law references, and legal keywords. These outputs feed directly into the retrieval tiers: Tier 1 uses the expanded query for tree navigation, while Tier 2 uses the seed entities to initialize PageRank and the intent label to adjust edge weights (1.0 for primary relations, 0.5 for others).

3.4.2. Tier 1: LLM-Guided Tree Traversal

Tier 1 takes the expanded query from Stage 0 and the document forest, and returns a ranked list of candidate articles with structural scores. It navigates the forest using LLM reasoning in three progressive loops: Loop 0 selects the top 5 documents by matching the query to domain groups via pre-computed keywords; Loop 1 narrows to the top 8 chapters using chapter summaries; Loop 2 identifies the top 7 articles per chapter from article summaries. General Provisions chapters are automatically included. Each loop produces a reasoning trace for interpretability.

3.4.3. Tier 2: Dual-Level Semantic Retrieval

Tier 2 takes the seed entities and intent label from Stage 0 together with the knowledge graph, and returns a separate ranked list of articles with semantic scores. It operates at two granularities.

The *low level* (entity-specific) computes four per-article signals. Keyphrase matching measures syllable n-gram overlap between query terms and KG entity names, aggregated per article as $\max + 0.3\ \ln(n+1)\,\bar{s}$ over n matched entities. Semantic similarity computes query–article cosine distance from sentence embeddings. Intent-aware PPR is seeded from entities whose keyphrase score ≥ 0.3. Concept matching propagates keyphrase hits to articles sharing the same ontology entity types, scoring by type-density. These are combined by weighted sum (w_{kp}=0.05, w_{sem}=0.20, w_{ppr}=0.25, w_{con}=0.20) and normalized to $[0,1]$.

The *high level* (conceptual) computes two signals. Theme matching embeds the query against pre-built theme clusters and maps source entities to articles. Hierarchy expansion then finds articles sharing entity types with theme matches (0.5 decay).

The two levels are fused: $s(a) = 0.70 \cdot s_{\text{low}}(a) + 0.15 \cdot s_{\text{theme}}(a) + 0.15 \cdot s_{\text{hier}}(a)$, summing to 1.0. Articles below threshold 0.1 are discarded.

The PPR component computes query-specific node importance on the knowledge graph [23]:

$$\text{PPR}(v) = (1-\alpha) \sum_{u \in N(v)} \frac{w(u,v) \cdot \text{PPR}(u)}{d(u)} + \alpha \cdot p(v) \tag{1}$$

where $\alpha = 0.15$ is the teleport probability, $w(u,v)$ is the edge weight adjusted by query intent, $d(u)$ is the weighted out-degree, and $p(v)$ is the personalization vector seeded from matched entities.

3.4.4. Tier 3: Semantic Bridge (RRF Fusion)

Tier 3 produces the final top-k articles through three steps: knowledge graph expansion, rank fusion, and adjacent enrichment.

First, seed articles from Tier 1 and Tier 2 are expanded through the knowledge graph's relation edges. Strong relations (*references*, *requires*, *depends_on*) are traversed both cross-chapter (up to 5 articles) and same-chapter (up to 2 articles), with a semantic relevance filter (cosine similarity ≥ 0.4) to suppress noise. This produces a third ranked list alongside Tier 1 and Tier 2.

The three lists are then merged using Reciprocal Rank Fusion [24]:

$$\mathrm{RRF}(d) = \sum_{s \in \{T1, T2, KG\}} w_s \cdot \frac{1}{k + \mathrm{rank}_s(d)} \tag{2}$$

where $k = 60$ and source weights $w_{T1} = 1.2$, $w_{T2} = 0.7$, $w_{KG} = 1.0$ are empirically set on a development set and shared across all domains. Tree traversal receives the highest weight as it provides the most reliable structural signal.

Finally, adjacent articles (± 2) from each fused result are added with decaying scores, as correct answers often lie in neighboring articles. The final ranked list is passed to an LLM for answer generation.

4. Implementation

The system is implemented in Python 3.11 with the following technical choices.

Document parsing. Legal texts are parsed with custom regex parsers that detect Vietnamese hierarchy markers ("Chuong", "Muc", "Dieu", "Khoan", "Diem") and build six-level trees. Scanned education PDFs (118 pages) are processed via LLM Vision OCR. Cross-references between documents are extracted via pattern matching on article citations.

Summary generation. Chapter-level and article-level summaries are generated via LLM (Claude 3.5 Haiku[2]), producing keyword-rich descriptions that Tier 1 tree traversal uses for branch selection without reading full article text.

Knowledge graph construction. Entity extraction and relation identification use Claude 3.5 Haiku via the Anthropic API, with one LLM call per article. Vietnamese slug normalization removes diacritics and collapses tonal variations to produce canonical entity identifiers. A multi-step entity resolver (exact match, synonym resolution, abbreviation expansion) merges equivalent entities across documents. The domain ontology defines 14 entity types using OWL classes with a 4-level hierarchy and Vietnamese label mappings.

Online query processing. Query analysis follows the pipeline described in Section 3.4. Tree traversal and article selection use Claude Sonnet 4 via the Anthropic API. Sentence embeddings are computed with paraphrase-multilingual-MiniLM-L12-v2[3], a 384-dimensional multilingual model from the Sentence-Transformers library [2]. Vector similarity search uses FAISS[4]. Personalized PageRank is computed iteratively (max 50 iterations, tolerance 10^{-6}, damping $\alpha = 0.15$). All LLM responses are cached in SQLite for reproducibility.

5. Experiments

5.1. Data Collection

Legal corpus. 18 Vietnamese legal and regulatory documents span three domains: enterprise law (Law 59/2020/QH14 and nine decrees, 77 chapters, 840 articles), traffic safety

[2] `https://www.anthropic.com/claude`
[3] `https://huggingface.co/sentence-transformers/paraphrase-multilingual-MiniLM-L12-v2`
[4] `https://github.com/facebookresearch/faiss`

law (Law 36/2024/QH15 and two penalty decrees), and education regulation (five post-graduate documents, 109 articles). Total: 949 articles.

Evaluation questions. 530 questions were collected from Vietnamese Facebook legal advice groups (400) and UIT student forums (130). A domain expert labeled each question with 1–8 relevant articles (average 2.3).

5.2. *Experimental Setup*

Table 1 summarizes the constructed knowledge graph.

Table 1. Knowledge graph statistics.

Entity Type	Count	Relation Type	Count	Structure	Count
Total Entities	5,936	Total Relations	5,963	Tree Nodes	9,226
Action	1,903	requires	1,377	Documents	13
Legal Term	1,632	applies_to	810	Chapters	77
Organization	500	has_authority	506	Articles	840
Legal Reference	451	references	434	Cross-refs	168
Other (9 types)	1,450	Other (52 types)	2,506		

Metrics. Since each query has expert-annotated relevant articles, we adopt standard retrieval metrics [11] that enable objective, reproducible evaluation against known ground truth, rather than LLM-as-judge scores used in open-ended generation benchmarks. Top-k hit rate $= \frac{1}{|Q|}\sum_q \mathbf{1}[\exists a \in R_q : \text{rank}(a) \leq k]$ measures the fraction of queries for which at least one relevant article appears in the top-k results, where R_q denotes the set of relevant articles for query q. Mean Reciprocal Rank is defined as MRR $= \frac{1}{|Q|}\sum_q (\min_{a \in R_q} \text{rank}(a))^{-1}$, averaging the reciprocal rank of the first relevant result. Recall@k, Precision@k, and NDCG@k [11] are also computed. Results are reported at $k \in \{1, 3, 5, 10, 20, 30\}$, with top-10 hit rate and MRR as the primary metrics. Top-10 hit rate is prioritized because in a RAG pipeline the LLM provides an additional validation layer over retrieved context, at least one correct article among ten candidates suffices for accurate answer generation while limiting noise that could induce hallucination. MRR complements top-10 hit rate by rewarding higher placement of relevant articles, reducing irrelevant context that the LLM must filter.

5.3. *Baselines*

Traditional: BM25 [10] and TF-IDF [11] with syllable-level tokenization. RAG systems: NaiveRAG [2] (dense retrieval, no reranking); LightRAG [4] (hybrid KG retrieval); RAPTOR [6] (collapsed-tree with LLM summaries); PageIndex [7] (two-step hierarchical tree search). All systems use Sonnet 4 for LLM calls and retrieve from the full corpus.

5.4. *Main Results*

Over three independent runs, the proposed system achieves $92.2 \pm 0.3\%$ top-10 hit rate (MRR 0.603), outperforming PageIndex by 7.4% and BM25 by 44.7% (Table 2). PageIndex benefits from tree navigation but misses cross-reference articles outside the traversal path. RAG baselines (NaiveRAG 32.8%, LightRAG 39.8%, RAPTOR 41.0%) underperform BM25 (47.5%): general-purpose embeddings lack Vietnamese legal vocabulary, generic KG extraction introduces noise, and RAPTOR's summaries lose article-level granularity. Our system overcomes these limitations via LLM-guided navigation and domain-specific knowledge graph.

Table 2. Legal domain hit rate (%) on 400 questions. Best in bold, second underlined.

Method	Top 5	Top 10	Top 20	Top 30	MRR
Traditional retrieval					
BM25	30.8	47.5	66.2	74.8	0.200
TF-IDF	29.2	45.8	64.5	74.8	0.207
Keyword	33.5	48.5	63.2	72.8	0.222
RAG systems					
NaiveRAG [2]	22.8	32.8	51.8	60.5	0.146
LightRAG [4]	25.5	39.8	47.8	49.2	0.173
RAPTOR [6]	25.2	41.0	54.5	62.3	0.170
PageIndex [7]	**76.2**	<u>84.8</u>	<u>90.2</u>	<u>91.0</u>	<u>0.559</u>
3-Tier (ours)	<u>74.5</u>	**92.2**	**94.8**	**95.0**	**0.603**

5.5. Multi-Domain Performance

Table 3 reports per-domain results, including education as a generalization test. The system outperforms all baselines across all three domains. Traffic law achieves the highest score (98.0%) due to direct violation-to-penalty mappings. Enterprise law is harder (86.5%) due to cross-decree references. Education (88.5%) shows the largest margin over PageIndex (+13.9%), where semantic retrieval captures terminology that tree navigation misses. The same retrieval pipeline and hyperparameters were used across all three domains without per-domain weight tuning.

Table 3. Per-domain top-10 hit rate (%) across three domains.

Method	Enterprise (197)	Traffic (203)	Education (130)
NaiveRAG	29.9	35.5	50.0
LightRAG	20.3	58.6	53.1
RAPTOR	18.5	63.5	53.8
BM25	31.5	63.1	70.0
TF-IDF	39.1	52.2	68.5
Keyword	20.3	75.9	56.2
PageIndex	<u>76.6</u>	<u>92.6</u>	<u>74.6</u>
3-Tier (ours)	**86.5**	**98.0**	**88.5**
Ours vs. PageIndex	*+9.9*	*+5.4*	*+13.9*

5.6. Analysis

PageIndex achieves a higher top-5 hit rate (76.2% vs. 74.5%), a precision–coverage tradeoff inherent in rank fusion: KG expansion articles occasionally displace tree results from positions 4–5 to 6–7 but recover cross-reference content that PageIndex never reaches, yielding a 17.7 pp gain from positions 6–10 compared to only 8.6 pp for PageIndex. In a RAG pipeline where the LLM filters ten candidates, this tradeoff is favorable.

Per-category top-10 hit rate ranges from 0% (household business, 4 queries outside the corpus) to 100% (speed penalties). A multi-hop example: "If a member borrows money to meet charter capital and later withdraws it, what are the penalties?" Tier 1 reaches Art. 47 (member obligations); PPR walks *requires* edges to Art. 16 Cl. 5 (prohibited acts) in a different chapter; RRF places both in the top 5. No baseline retrieves this at top 30. Of the 31 missed queries (7.8%), 27 involve enterprise law: cross-decree dependencies (6), out-of-corpus articles (4), and multi-article queries requiring >2 articles (14), mostly linked by implicit legal reasoning chains rather than explicit cross-references.

6. Conclusion

This paper presented a three-tier knowledge graph RAG architecture for Vietnamese legal question answering, combining structural tree traversal, semantic graph search with intent-aware Personalized PageRank, and Reciprocal Rank Fusion. Experiments across three domains demonstrate consistent superiority over all baselines using the same pipeline and hyperparameters without per-domain tuning. Limitations include Tier 1 latency (around 7s per query due to multiple LLM calls) and lower performance on enterprise queries requiring multi-document cross-references. Future work includes learned weight optimization to replace empirical tuning, extension to additional regulatory domains, and multi-turn conversational support.

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References

[1] Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, Sebastian Riedel, and Douwe Kiela. Retrieval-augmented generation for knowledge-intensive NLP tasks. In *Advances in Neural Information Processing Systems 33 (NeurIPS)*, pages 9459–9474, 2020.

[2] Nils Reimers and Iryna Gurevych. Sentence-BERT: Sentence embeddings using siamese BERT-networks. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 3982–3992, 2019.

[3] Wenqi Fan, Yujuan Ding, Liangbo Ning, Shijie Wang, Hengyun Li, Dawei Yin, Tat-Seng Chua, and Qing Li. A survey on RAG meeting LLMs: Towards retrieval-augmented large language models. In *Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD)*, pages 6491–6501, 2024.

[4] Zirui Guo et al. LightRAG: Simple and fast retrieval-augmented generation. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2025.

[5] Darren Edge, Ha Trinh, Newman Cheng, Joshua Bradley, Alex Chao, Apurva Mody, Steven Truitt, and Jonathan Larson. From local to global: A graph RAG approach to query-focused summarization. Technical report, Microsoft Research, 2024.

[6] Parth Sarthi, Salman Abdullah, Aditi Tuli, Shubh Khanna, Anna Goldie, and Christopher D. Manning. RAPTOR: Recursive abstractive processing for tree-organized retrieval. In *International Conference on Learning Representations (ICLR)*, 2024.

[7] Mingtian Zhang, Yu Tang, and PageIndex Team. PageIndex: Next-generation vectorless, reasoning-based RAG. *PageIndex Blog*, September 2025. `https://pageindex.ai/blog/pageindex-intro`.

[8] Thi Thanh Sang Nguyen, Hoang Dung Nguyen, Van Dung Pham, Duc Thin Tran, and Ali Selamat. Legal-onto: An ontology-based model for representing the knowledge of a legal document. In *Proceedings of the 17th International Conference on Evaluation of Novel Approaches to Software Engineering (ENASE)*, pages 426–434, 2022.

[9] Giovanni Sartor et al. *Approaches to Legal Ontologies: Theories, Domains, Methodologies*. Springer, Dordrecht, 2011.

[10] Stephen Robertson and Hugo Zaragoza. The probabilistic relevance framework: BM25 and beyond. *Foundations and Trends in Information Retrieval*, 3(4):333–389, 2009.

[11] Christopher D. Manning, Prabhakar Raghavan, and Hinrich Schütze. *Introduction to Information Retrieval*. Cambridge University Press, 2008.

[12] Vladimir Karpukhin, Barlas Oğuz, Sewon Min, Patrick Lewis, Ledell Wu, Sergey Edunov, Danqi Chen, and Wen-tau Yih. Dense passage retrieval for open-domain question answering. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 6769–6781, 2020.

[13] Akari Asai, Zeqiu Wu, Yizhong Wang, Avirup Sil, and Hannaneh Hajishirzi. Self-RAG: Learning to retrieve, generate, and critique through self-reflection. In *Proceedings of the International Conference on Learning Representations (ICLR)*, 2024.

[14] Luyu Gao, Xueguang Ma, Jimmy Lin, and Jamie Callan. Precise zero-shot dense retrieval without relevance labels. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (ACL)*, pages 1762–1777, 2023.

[15] Ilias Chalkidis, Manos Fergadiotis, Prodromos Malakasiotis, Nikolaos Aletras, and Ion Androutsopoulos. LEGAL-BERT: The muppets straight out of law school. In *Findings of the Association for Computational Linguistics: EMNLP 2020*, pages 2898–2904, 2020.

[16] Dat Quoc Nguyen and Anh Tuan Nguyen. PhoBERT: Pre-trained language models for Vietnamese. In *Findings of the Association for Computational Linguistics: EMNLP 2020*, pages 1037–1042, 2020.

[17] Thanh Vu, Dat Quoc Nguyen, Dai Quoc Nguyen, Mark Dras, and Mark Johnson. VnCoreNLP: A Vietnamese natural language processing toolkit. In *Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics: Demonstrations (NAACL)*, pages 56–60, 2018.

[18] Hung Ngo, Thi Nguyen, Dung Nguyen, and Minh Pham. AimeLaw at ALQAC 2021: Enriching neural network models with legal-domain knowledge. In *Proceedings of the 13th International Conference on Knowledge and Systems Engineering (KSE)*. IEEE, 2021.

[19] Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. BERT: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (NAACL-HLT)*, pages 4171–4186, 2019.

[20] Nhon V. Do, Hoang Dung Nguyen, and Ali Selamat. Knowledge-based model of expert systems using rela-model. *International Journal of Software Engineering and Knowledge Engineering*, 28(8):1047–1090, 2018.

[21] Haoxi Zhong, Chaojun Xiao, Cunchao Tu, Tianyang Zhang, Zhiyuan Liu, and Maosong Sun. How does NLP benefit legal system: A summary of legal artificial intelligence. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics (ACL)*, pages 5218–5230, 2020.

[22] Jinqi Lai, Wensheng Gan, Jiayang Wu, Zhenlian Qi, and Philip S. Yu. Large language models in law: A survey. *AI Open*, 5:181–196, 2024.

[23] Taher H. Haveliwala. Topic-sensitive PageRank. In *Proceedings of the 11th International Conference on World Wide Web (WWW)*, pages 517–526, 2002.

[24] Gordon V. Cormack, Charles L.A. Clarke, and Stefan Buettcher. Reciprocal rank fusion outperforms condorcet and individual rank learning methods. In *Proceedings of the 32nd International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR)*, pages 758–759, 2009.